CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) — CHAPTER 7

SHARE AND REVIEW A SPREADSHEET — STRUCTURED NOTEBOOK & REVISION NOTES

Syllabus Sub-Unit 4 Complete Theory • Sharing Mode • Conflict Resolution • Track Changes & Comments • Merging & Comparing

---

1. Document Collaboration & User Registration

Concept of Collaboration: When an organization deals with large volumes of data spread across multiple locations or departments, several users often need to work on the exact same spreadsheet file simultaneously (e.g., regional correspondents entering product survey responses or finance officers compiling quarterly budget figures).

- **User Registration Purpose:** Before collaborating or auditing changes in LibreOffice, each user must register their identity. This ensures that every edit, cell change, and comment is automatically attributed to the correct user with an accurate author name and timestamp.
- **Step-by-Step Registration Procedure:**
  1. Choose **Tools** → **Options...** from the Menu bar (or press **Alt + F12**).
  2. In the left panel of the Options window, expand the **LibreOffice** group and select **User Data**.
  3. In the right pane, fill in: *First/Last Name/Initials*, *Company*, *City*, *Country*, and *Title/Position*.
  4. Ensure the checkbox **'Use data for document properties'** is checked.
  5. Click **Apply** and **OK**.

2. Setting Up, Opening & Saving a Shared Spreadsheet

Definition: Sharing a spreadsheet allows multiple users to open, view, edit, and review the same workbook concurrently over a local network or shared cloud folder.

A. Setting Up a Spreadsheet for Sharing (Step-by-Step):

Step 1: Open the spreadsheet document you want to share.

Step 2: Choose **Tools** → **Share Spreadsheet...** from the Menu bar. The *Share Document* dialog opens.

Step 3: Check the checkbox labelled **'Share this spreadsheet with other users'**. *(Unchecking this box disables sharing and returns the file to exclusive single-user access mode).*

Step 4: The bottom pane displays the list of users currently accessing the file with login timestamps.

Step 5: Click **OK**. If there are unsaved edits, Calc prompts that the file must be saved to begin sharing. Click **Yes**.

Step 6: Visual Title Bar Confirmation: Once sharing is active, Calc displays the word **(shared)** immediately after the filename on the Title Bar (e.g., CompanyBudget.ods (shared) - LibreOffice Calc).

B. Opening a Shared Spreadsheet:

Open via **File** → **Open** (or **Ctrl + O**). Calc displays an informational banner explaining that the workbook is in shared mode and that certain editing features are unavailable. Click **OK** to proceed into shared mode.

C. Features Allowed vs. Disabled in Shared Workbooks:

Feature Category	Allowed / Supported Features	Disabled / Prohibited Features in Shared Mode
Data Entry & Edits	Entering/editing text, numbers, dates, and formulas; deleting cell contents.	Conditional formatting; Data Validation (Validity); Define Range; Subtotals; Consolidate; Multiple Operations.
Formatting	Direct cell text formatting (fonts, font size, bold/italic).	**Merging cells (Format → Merge Cells)**; formatting print ranges.
Insert Objects	Viewing existing charts, images, and drawings.	**Inserting pictures/graphs**; inserting charts, OLE objects, floating frames, media, Fontwork, signature lines.
Sheet Architecture	Switching between existing sheets; viewing tabs.	Insert Sheet from File; Rename Sheet; Delete Sheet; Move/Copy Sheet; Sheet Tab Color.
Review & Tracking	**Merge Document** (merging changes from another file).	Edit → Track Changes (Record, Show, Manage are disabled in shared mode); inserting comments.

D. Saving a Shared Spreadsheet & Conflict Resolution:

- **Case 1: No Conflicts (Non-Overlapping Edits):** If multiple users edit different cell ranges (e.g., User 1 edits *A10:F30* while User 2 edits *A55:E80*), the file saves normally. Calc updates the sheet with a message: *'Your spreadsheet has been updated with changes saved by other users.'* Modified cells saved by other users are highlighted with a **red border**.
- **Case 2: Conflicting Changes (Overlapping Edits):** If two or more users modify the *exact same cell* with conflicting values, the **Resolve Conflicts** dialog box appears automatically upon saving.
- **Conflict Resolution Action Options:**
  - **Keep Mine:** Retains your newly entered value and overwrites the conflicting change.
  - **Keep Other:** Discards your change and retains the value saved by the other user.
  - **Keep All Mine / Keep All Others:** Resolves all remaining cell conflicts at once.
  - *File Locking Rule:* While a user is resolving conflicts, the file is temporarily locked; other users cannot save and are notified that a merge-in is in progress.

A.S. NOTES IT-402

3. Recording Changes in a Spreadsheet (Audit & Review Mode)

Definition: Recording Changes is a specialized auditing mode in Calc where all insertions, deletions, and cell edits are logged as provisional modifications. The original author or manager can later review each edit and decide whether to accept or reject it.

- **Prerequisites Before Enabling Record Changes:**
  1. **Sharing Must Be Turned Off:** Record Changes cannot be activated while a spreadsheet is in shared mode. Users must first disable sharing via Tools → Share Spreadsheet... (uncheck the box) and save the file.
  2. **Confirm Sharing is Disabled:** Check the Title Bar to verify that the tag **(shared)** is no longer present.
- **Step-by-Step Procedure to Record Changes:**
  1. Open the unshared spreadsheet document.
  2. Choose **Edit** → **Track Changes** → **Record** from the Menu bar (or press **Ctrl + Shift + C**). A checkmark appears next to *Record*.
  3. Perform your data entries, updates, or corrections in the worksheet cells.
  4. **Visual Cell Indicator:** Calc draws a **coloured border with a dot in the upper left-hand corner** around every modified cell.
  5. **Hover Tooltip:** Hovering over the cell displays an automated tooltip showing who made the change, date, time, and previous value (e.g., *'pace2, 29/07/2026 15:30:44: Cell C7 changed from 136 to 236'*).
  6. *Password Protection:* Choose **Edit** → **Track Changes** → **Protect...** and enter a password to prevent unauthorized reviewers from disabling tracking or accepting their own edits.

4. Adding, Editing and Formatting Comments in Changed Cells

In addition to the automatic system comment logging old and new values, reviewers can attach explanatory comments to justify their modifications.

- **Adding an Explanatory Comment:** Select the changed cell → Choose **Edit** → **Track Changes** → **Comment...** (or press **Ctrl + Alt + C**) → Type your note in the Text box → Use arrow buttons to step through other changed cells → Click **OK**.
- **Editing an Existing Change-Comment:** Select the cell → Choose **Edit** → **Track Changes** → **Comment...** → Modify text → Click **OK**.
- **Formatting a Comment Box:** Right-click the cell containing the comment → Select **Edit Comment** → Highlight text inside the comment box → Use the **Formatting Toolbar** to customize font, size, color, background highlight, or alignment.

5. Reviewing, Filtering & Accepting/Rejecting Changes

Once reviewer edits are complete, the owner reviews the modifications to finalize the spreadsheet.

- **A. Filtering Visible Changes (Show Changes Dialog):**
  - Choose **Edit** → **Track Changes** → **Show...**.
  - Filter visible markups by: **Date** (time intervals), **Author** (specific reviewers), **Range** (specific cell blocks like *Sheet1.B4:E20*), **Comment** (matching text strings), and checkboxes for **Show accepted changes** / **Show rejected changes**.
- **B. Managing Changes (Accepting or Rejecting Edits):**
  1. Open the reviewed spreadsheet and choose **Edit** → **Track Changes** → **Manage...**.
  2. The *Manage Changes* dialog displays a list of all pending changes with *Action*, *Position*, *Author*, *Date*, and *Comment*.
  3. Click any entry to highlight the corresponding cell in the worksheet.
  4. Click **Accept** (permanently applies the edit), **Reject** (reverts to original value), **Accept All** (approves all edits), or **Reject All** (discards all edits).
  5. Click **Close** and save the finalized document.

6. Merging and Comparing Spreadsheets

In distributed environments, multiple team members often edit copies of the same file separately. Calc provides two methods to consolidate them:

- **A. Merging Documents (When Track Changes Was Turned On):**
  - *Why Merge?* Combines recorded changes from multiple reviewed copies into the master original sheet at once, allowing the author to review all contributors' proposals together.
  - *Prerequisite:* All reviewed copies **must have had Record Changes enabled** during editing.
  - *Step 1:* Open the master original document (e.g., *Budget_Original.ods*).
  - *Step 2:* Choose **Edit** → **Track Changes** → **Merge Document...**.
  - *Step 3:* Select the reviewed file (e.g., *Budget_Raj.ods*) and click **Open**.
  - *Step 4:* The *Manage Changes* dialog opens displaying merged edits. If multiple reviewers modified the exact same cell, Calc displays a **plus (+) sign** next to the entry; click the **+** to expand and inspect each reviewer's proposed value.
  - *Step 5:* Accept or reject edits as desired, repeat for other copies (e.g., *Budget_Meera.ods*), and save.
- **B. Comparing Documents (When Reviewers Forgot to Turn On Track Changes):**
  - *Why Compare?* If a reviewer edited a copy *without recording changes*, Calc compares the modified version against the original file to identify all discrepancies automatically.
  - *Step 1:* Open the edited file (e.g., *Rev_Bill.ods*).
  - *Step 2:* Choose **Edit** → **Track Changes** → **Compare Document...** from the Menu bar.
  - *Step 3:* Select the original document (e.g., *Bill.ods*) and click **Open**.
  - *Step 4:* Calc compares the files and opens the *Manage Changes* dialog, categorizing differences into:
    - **Inserted:** Data found in the edited file but absent in the original.
    - **Deleted:** Data found in the original file but missing from the edited file.
    - **Changed:** Cells where contents differ between the two versions.
  - *Step 5:* Review each difference, click **Accept** or **Reject**, and save the updated file.

DAV GVM IT-402